Concrete Frontend Architecture & Execution Plan

Target Aesthetic: Linear / Cloudflare / Snyk / Datadog (High-density, authoritative, zero cyberpunk/neon slop).
Design Standard: Strict adherence to `taste-skill` (`design-taste-frontend` v2.5.0).
Owner: Member 3 (Frontend & UI/UX Lead).

1. Visual Design System & Design Tokens

Color Palette (Subdued, Functional, Authoritative)

No neon glowing borders or matrix greens. We use a grounded neutral zinc base with functional status accents:

Token	Hex Value	Usage
Ground (Background)	`#09090b` (zinc-950)	Full page viewport ground
Surface (Cards/Panels)	`#121215` / `#18181b` (zinc-900)	Bento panels, app-frame container
Subtle Border	`#27272a` (zinc-800)	1px clean borders for cards and inputs
Border Active / Hover	`#3f3f46` (zinc-700)	Subtle 1-2px lift and crisp boundary
Primary Typography	`#f4f4f5` (zinc-100)	Headlines, primary metrics
Muted Typography	`#71717a` (zinc-500)	Labels, descriptions, timestamps
Status Safe / Valid	`#10b981` (emerald-500)	Valid X.509 chain, clean domain, safe pass
Status Warning	`#f59e0b` (amber-500)	Missing HSTS/CSP, untrusted root, self-signed
Status Critical Threat	`#e11d48` (rose-600)	eMudhra weak scalar key ($d = 4$d = 4), phishing clone
Status Informational	`#0284c7` (sky-600)	CT log entries, STIX identifiers, DNS records

Typography

- **UI Interface Font:** Geist Sans / Inter (`font-sans`), tight tracking (`tracking-tight`).
- **Cryptographic & Code Data:** Geist Mono / JetBrains Mono (`font-mono`) for public keys, modulus, scalar integers, PEM certs, SHA-256 fingerprints, and DNS records.

Micro-Interactions (Rule 5 of `taste-skill`)

- Cards lift **1–2px** on hover with a crisp border highlight (`hover:border-zinc-700 transition-all duration-200`).
- Buttons have physical click states: `active:scale-[0.98]`.
- Zero cartoon bounce animations, zero auto-spinning carousels.

2. Screen-by-Screen Layout (Bento Grid Architecture)

Following the rule **"One section per screen at 1440x900 viewport"**, the application has 4 distinct, cohesive screens:

```
+-------------------------------------------------------------------------------
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|                                    APPLICATION MAP
|
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|  SCREEN 1: The Threat Command Center & Parallel Inspection Pipeline (The Hero)
|
|  SCREEN 2: The Cryptographic Trust & eMudhra Auditor (The Deep Differentiator)
|
|  SCREEN 3: Phishing Heuristics & Web Application Security Posture
|
|  SCREEN 4: Collaborative Threat Intelligence Radar & Live Browser Guard
Simulation     |
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```

Screen 1: The Command Console & Parallel Inspection Pipeline (The Hero)

- **Top Bar:**
  - Logo: `K9-SENTINEL` (monospaced badge `v1.0.4-PROD`).

    - Real-time indicator: `SYSTEM: ARMED` (subtle emerald dot).
    - Quick action: `Export STIX 2.1` and `View API Docs`.
- **Universal Target Scanner Input:**
    - High-precision URL bar with placeholder: `Enter target URL or domain (e.g., https://secure-bank.example.com)...`
    - Deep Scan action button with instantaneous hotkey (`Enter` or `Cmd+K`).
- **Preset Scenario Pills (Crucial for Live Demos):**
    - Pill 1: 🚨 `eMudhra Flawed CA (d=4)`
    - Pill 2: ⚠️ `PayPal Phishing Clone`
    - Pill 3: 🔓 `Insecure University Portal`
    - Pill 4: ✅ `Legitimate Domain (google.com)`
- **The Inspection Pipeline (Based on `lanes` layout from `taste-skill`):** An interactive horizontal diagram showing the target domain entering a central dispatcher and feeding 4 parallel lanes:
    - *Lane 1:* Cryptographic Scalar & Chain Audit
    - *Lane 2:* Typosquatting & Homograph Verification
    - *Lane 3:* HTTP Security Posture Audit
    - *Lane 4:* Collaborative Threat Feeds
- **Primary Threat Verdict Card:**
    - Radial / Circular Score Gauge (0–100 Risk Score).
    - Decisive Status Badge: `CRITICAL THREAT` / `SUSPICIOUS` / `VERIFIED SAFE`.
    - Executive summary line (e.g., *"Immediate impersonation detected: Rogue intermediate CA certificate with trivially small private scalar d=4."*).

Screen 2: The Cryptographic Trust & eMudhra Auditor (The Judge Magnet)

This screen proves mathematical depth and avoids AI slop by displaying real cryptographic primitives:

- **Interactive X.509 Trust Chain Hierarchy:**
    - Root CA (e.g. `Global Root Authority`) → →
    - Intermediate CA (Highlighted with an amber/red warning border: `eMudhra Sub-CA`) → →
    - Leaf Domain Certificate (`*.target-bank.com`).

- **The eMudhra Flawed Key Inspector (Bug 2083449):**
  - **Curve Details:** NIST P-384 / P-256.
  - **Public Point Coordinates:** $Q = (x, y)$Q = (x, y) rendered in crisp monospaced blocks with copy button.
  - **Scalar Verification Metric:** Candidate scalar test $Q' = d \cdot G$Q′ = d ⋅ G.
  - **Discovered Private Scalar:** `d = 4` (Highlighted in ruby text: *"CRITICAL: Low-entropy private key derived in 0.04 ms"*).
- **ROCA & RSA Key Quality Audit:**
  - Key Length: `2048-bit` / `4096-bit` / `1024-bit (FAIL)`.
  - ROCA Vulnerability Check: Pass / Fail.
  - Signature Algorithm: SHA-256 with RSA (flags deprecated SHA-1/MD5).
- **Certificate Transparency (CT) Timeline:**
  - Chronological list of certificates issued for this domain logged on public CT logs (crt.sh).
  - Highlights sudden re-issuances or unauthorized certificate velocity.

Screen 3: Phishing Heuristics & Application Security Posture

- **Left Panel: Phishing & Impersonation Heuristics (2/3 width):**
  - **Typosquatting & Homograph Comparison:**
    - Target Domain vs. Legitimate Domain side-by-side (`paypa1.com` vs `paypal.com`).
    - Levenshtein Distance Badge: `Distance: 1 (Targeting PayPal Inc.)`.
    - Punycode / Cyrillic Homograph Flag (e.g. `xn--...` substituting `a` for `a`).
  - **DOM Credential Harvesting Scanner:**
    - Password / Card input detection badge.
    - Form Action Audit: Flags cross-origin exfiltration (e.g. Form on `domainA.com` posts credentials to `evil-drop.net/steal.php`).
  - **Deceptive Urgency NLP Heuristics:**
    - Flags keywords detected in DOM: *"Urgent KYC Verification"*, *"Account Suspended within 24h"*.
- **Right Panel: Web Application Security Posture (1/3 width):**

    - Security Headers Grade: `A+`, `A`, `B`, `C`, `F`.
    - Checklist of critical headers:
        - `Strict-Transport-Security` (HSTS)
        - `Content-Security-Policy` (CSP)
        - `X-Frame-Options` (Clickjacking defense)
        - `X-Content-Type-Options: nosniff`
    - Exposed Sensitive Files Probe:
        - Status for `/.git/HEAD`, `/.env`, `/wp-config.php.bak`, `/swagger.json`.

Screen 4: Collaborative Threat Intelligence & Browser Guard Demo

- **Collaborative Threat Radar (Live Community Feed):**
    - Real-time stream of recently submitted suspicious links.
    - Interactive Community Voting:
        - `[ Confirm Phishing ]` (+1 Reputation)
        - `[ Flag False Positive ]` (-1 Reputation)
    - Community Consensus Bar (e.g. 94% Consensus based on 28 security researcher votes).
    - One-click **STIX 2.1 / MISP JSON Export** button for enterprise SIEM ingestion.
- **Interactive Browser Extension Simulator:**
    - An authentic browser window mockup (URL bar, navigation buttons, viewport).
    - Displays what the user sees when navigating to a flagged malicious site:
        - A crisp, urgent warning banner: *"K9-Sentinel Shield: Navigation Intercepted"*.
        - Details: *"This site impersonates PayPal and uses an insecure eMudhra TLS certificate."*
        - Actions: `[ Go Back to Safety ]` or `[ Proceed at Own Risk ]`.

3. Frontend Component Structure (`frontend/src/`)

```
frontend/src/
├── app/
│   ├── globals.css                # Zinc theme, custom scrollbars, subtle 1px
border utilities
│   ├── layout.tsx                 # Root layout with Geist/Inter font
│   └── page.tsx                   # Main orchestrator mounting the 4 screens
│
├── components/
│   ├── Navbar.tsx                 # Minimal top navigation & system status pill
│   ├── HeroScanner.tsx            # URL input bar + Quick scenario selector
│   ├── PipelineVisualizer.tsx     # 4-lane parallel analysis diagram (taste-skill
spec)
│   ├── RiskGauge.tsx              # 0-100 Radial Threat Index gauge
│   │
│   ├── crypto/
│   │   ├── CertificateTree.tsx    # Root -> Intermediate -> Leaf visual hierarchy
│   │   ├── EmudhraAuditCard.tsx   # P-384 scalar d=4 discovery console
│   │   └── CTLogTimeline.tsx      # Certificate Transparency chronological feed
│   │
│   ├── heuristics/
│   │   ├── TyposquattingCard.tsx  # Brand diff visualizer with Levenshtein
distance
│   │   ├── FormHarvestingCard.tsx # Cross-origin form exfiltration alert
│   │   └── HeaderScorecard.tsx    # HSTS, CSP, XFO audit badges
│   │
│   ├── collaborative/
│   │   ├── ThreatRadarFeed.tsx    # Live incident stream with upvote/downvote
│   │   └── STIXExportModal.tsx    # STIX 2.1 JSON inspector & download
│   │
│   └── demo/
│       └── BrowserGuardMockup.tsx # Interactive Chrome browser extension
simulation
│
├── lib/
│   ├── mockData.ts                # Full, realistic mock data for all 4 scenarios
│   └── api.ts                     # Fetch client to connect to FastAPI backend
│
└── types/
    └── api.ts                     # TypeScript definitions for scan results and
threat feeds
```

4. Execution Roadmap for Member 3

Step	Time	Task	Deliverable
Step 1	Hour 1	Initialize Next.js 15, install Lucide icons & Tailwind, configure zinc theme	Clean working dev server at `localhost:3000`
Step 2	Hour 2	Build `Navbar.tsx`, `HeroScanner.tsx`, and `RiskGauge.tsx` with preset scenario buttons	Screen 1 fully interactive with toggleable presets
Step 3	Hour 3	Build `CertificateTree.tsx` and `EmudhraAuditCard.tsx` (the eMudhra inspector)	Screen 2 showing X.509 chain and scalar $d = 4$d = 4 alert
Step 4	Hour 4	Build `TyposquattingCard.tsx`, `FormHarvestingCard.tsx`, and `HeaderScorecard.tsx`	Screen 3 displaying heuristics and security posture
Step 5	Hour 5	Build `ThreatRadarFeed.tsx` (voting buttons) and `BrowserGuardMockup.tsx`	Screen 4 collaborative feed and extension demo
Step 6	Hour 6	Connect `lib/api.ts` to backend REST endpoints; run `npm run build` to verify zero errors	Production-ready frontend ready for Vercel deployment

5. Ready-to-Use Mock Data for Member 3

Member 3 does NOT need to wait for backend deployment to start building. The 4 demo scenarios are pre-defined:

1. `SCENARIO_EMUDHRA`: Fails cryptographic test (P-384, scalar $d = 4$d = 4, intermediate revoked in OneCRL).
2. `SCENARIO_PAYPAL_PHISH`: Fails brand check (`paypa1-secure-login.com`, harvesting password to foreign IP).
3. `SCENARIO_INSECURE_APP`: Missing HSTS/CSP, exposed `/.git/HEAD`.
4. `SCENARIO_CLEAN_GOOGLE`: 100% clean, valid Google Trust Services CA, Grade A+ headers.